

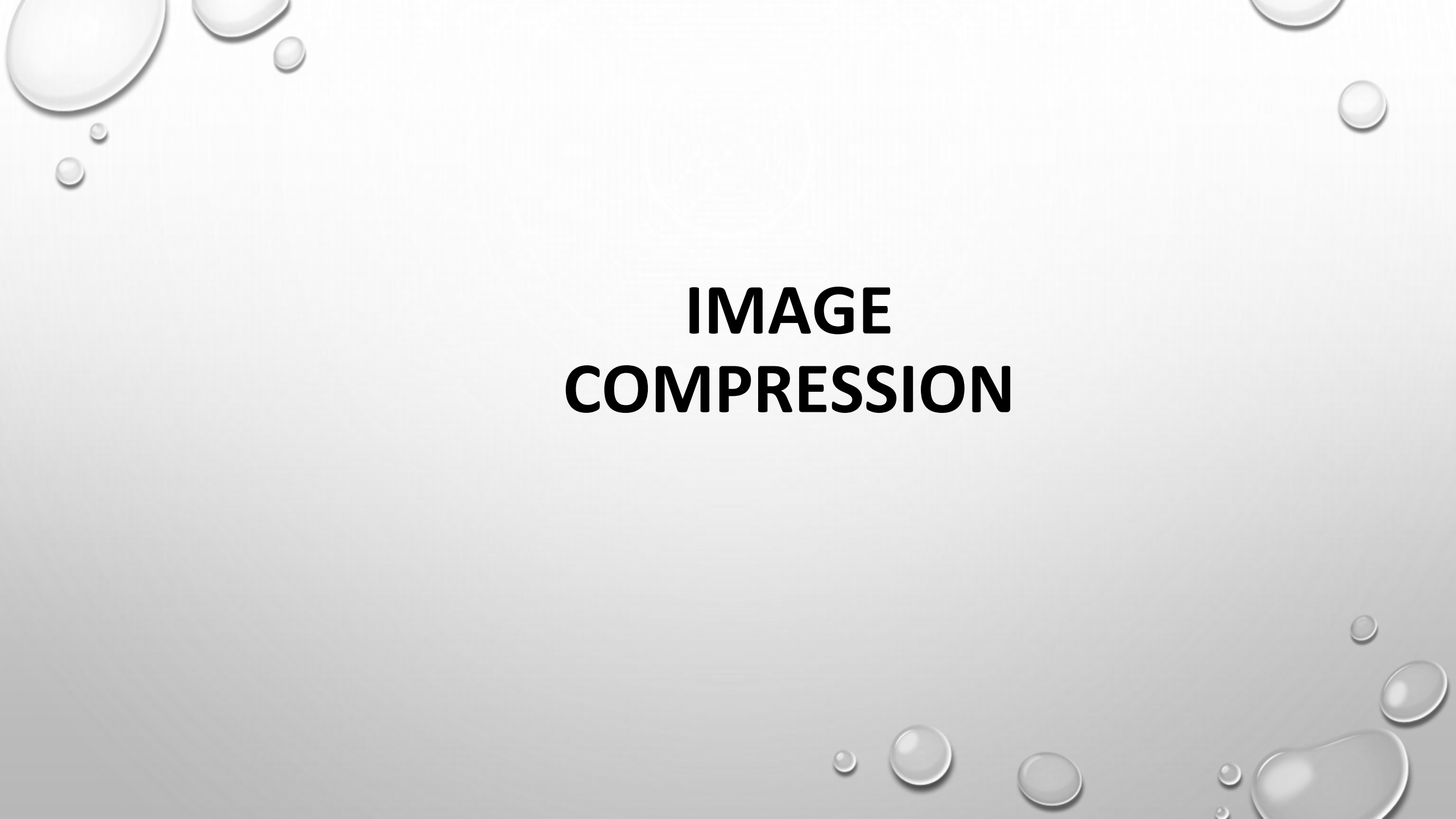
The background is a light gray gradient. In the top-left and bottom-right corners, there are several realistic water droplets of various sizes, rendered with highlights and shadows to give them a 3D appearance. The text "IMAGE COMPRESSION" is centered in the middle of the slide.

IMAGE COMPRESSION

FUNDAMENTALS

- ❑ **Data compression:** it refers to the process of reducing the amount of data required to represent a given quantity of information.

Data vs information

- ❑ **Data and information are not synonymous terms!**
- ❑ **Data** is the means by which **information** is conveyed.

DATA VS INFORMATION (CONT'D)

- ❑ The same **information** can be represented by different amount of **data** – for example:

Example 1: your brother, rohit, will meet you at IGI airport in new delhi at 5 minutes past 6:00 pm tomorrow night

Example 2: your brother will meet you at IGI airport at 5 minutes past 6:00 pm tomorrow night

Example 3: rohit will meet you at IGI at 6:00 pm tomorrow night

DATA ≠ INFORMATION

- ❑ Data compression aims to reduce the amount of data while preserving as much information as possible.
- ❑ **Because various amount of data can be used to represent the same amount of information,** representations that contain irrelevant or repeated information are said to contain *redundant data*.

INTRODUCTION

- ❑ **Image compression:** it is the art & science of reducing the amount of data required to represent an image.
- ❑ The number of images that are compressed and decompressed daily is staggering, and the **compressions and decompressions themselves are virtually invisible to the user.**
- ❑ Anyone who owns a digital camera, surfs the web, or watches the latest hollywood movies (on dvds or yify torrents) benefits from the compression algorithms

DATA VS INFORMATION EXAMPLE

- ❑ In terms of an image, information corresponds to the object captured in the image, their shape, their color etc.
- ❑ The different ways to represent this information can be either:
 - **Through pixel intensities (spatial domain) (basic), or**
 - **Through extracting features of that image (higher level of information e.g. edge image.. Edge out of prev image), or**
 - **Through simply noting the frequencies of the intensity values existing in the image along with their location (frequency domain), etc.**

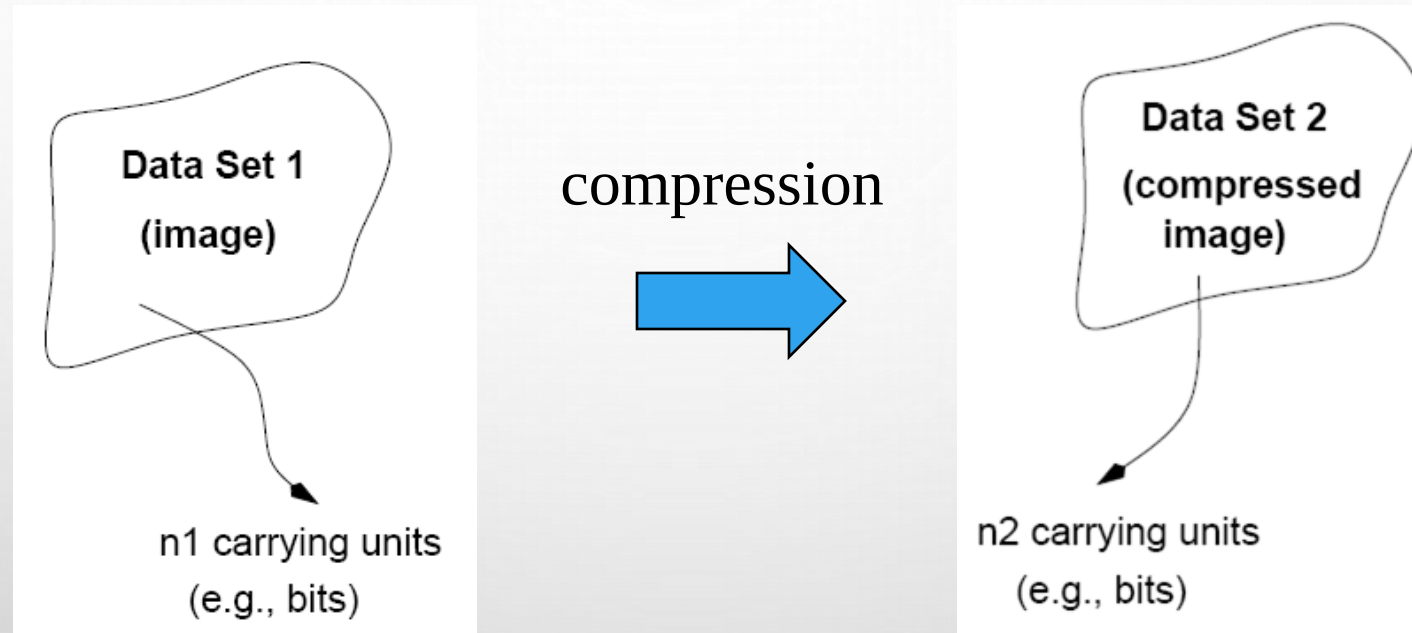
COMPRESSION RATIO

- ❑ *Image compression* involves **reducing the size** of image data files, while ***retaining necessary information***
- ❑ Retaining necessary information depends upon the **application**
- ❑ The reduced file created by the compression process is called the ***compressed file*** and is used to reconstruct the image, resulting in ***the decompressed image***
- ❑ The original image, before any compression is performed, is called the ***uncompressed image file***
- ❑ The ratio of the original, uncompressed image file and the compressed file is referred to as the ***compression ratio***

Data Redundancy

- Data that provide no relevant information=*redundant data* or *redundancy*.
- Image compression techniques can be designed by reducing or eliminating the **Data Redundancy**
- Image coding or compression has a goal to reduce the amount of data by reducing the amount of redundancy.

COMPRESSION RATIO



Let n_1 and n_2 denote the number of bits in two representations of the same information

Data Redundancy

Let n_1 and n_2 refer to amounts of data in two data sets that carry the same information

Compression ratio: $C_R = \frac{n_1}{n_2}$

Relative data redundancy: $R_D = 1 - \frac{1}{C_R} = 1 - \frac{n_2}{n_1}$
(of the first data set, n_1)

- if $n_1 = n_2$, $C_R = 1$ and $R_D = 0$, relative to the second data set, the first set contains no redundant data
- if $n_1 \gg n_2$, $C_R \rightarrow \infty$, and $R_D \rightarrow 1$, relative to the second data set, the first set contains highly redundant data
- if $n_1 \ll n_2$, $C_R \rightarrow 0$, and $R_D \rightarrow -\infty$, relative to the second data set, the first set is highly compressed

$C_R = 10$ means 90% of the data in the first data set is redundant

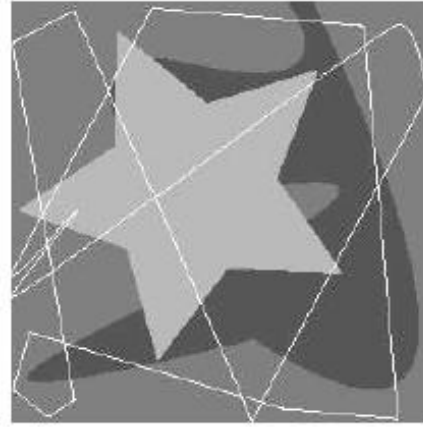
TYPES OF DATA REDUNDANCY

- ❑ Three principal types of data redundancies that can be identified and exploited in digital images
 1. Coding redundancy
 2. Spatial or temporal (interpixel) redundancy
 3. Psychovisual redundancy (irrelevant information)

- ❑ Data compression attempts to reduce one or more of these redundancy types.

CODING REDUNDANCY

The 8-bit codes that are used to represent the intensities in most 2-D intensity arrays contain more bits than are needed to represent the intensities.



Because, we go for uniform coding, in general.

We can solve the problem by variable length coding.

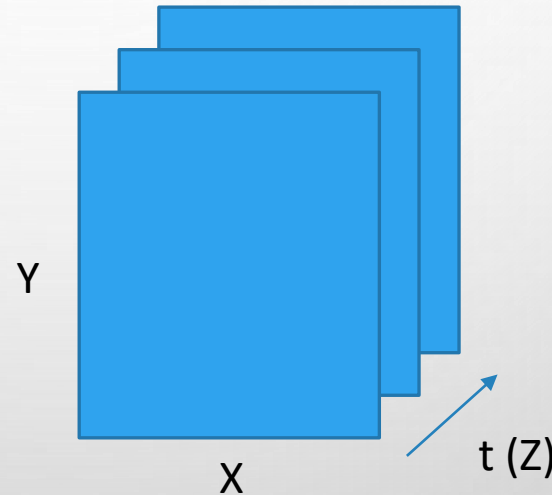
r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	10000000	8	1	1
$r_{186} = 186$	0.25	11000100	8	000	3
$r_{255} = 255$	0.03	11111111	8	001	3
r_k for $k \neq 87, 128, 186, 255$	0	—	8	—	0

SPATIAL OR TEMPORAL (INTERPIXEL) REDUNDANCY

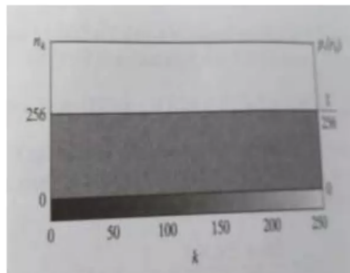
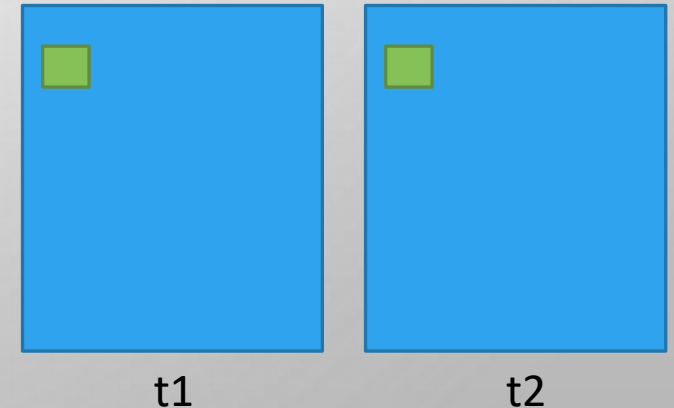
Interpixel redundancy implies that pixel values are correlated (i.e., A pixel value can be reasonably predicted by its neighbors).

Temporal

Related to 3D (videos)



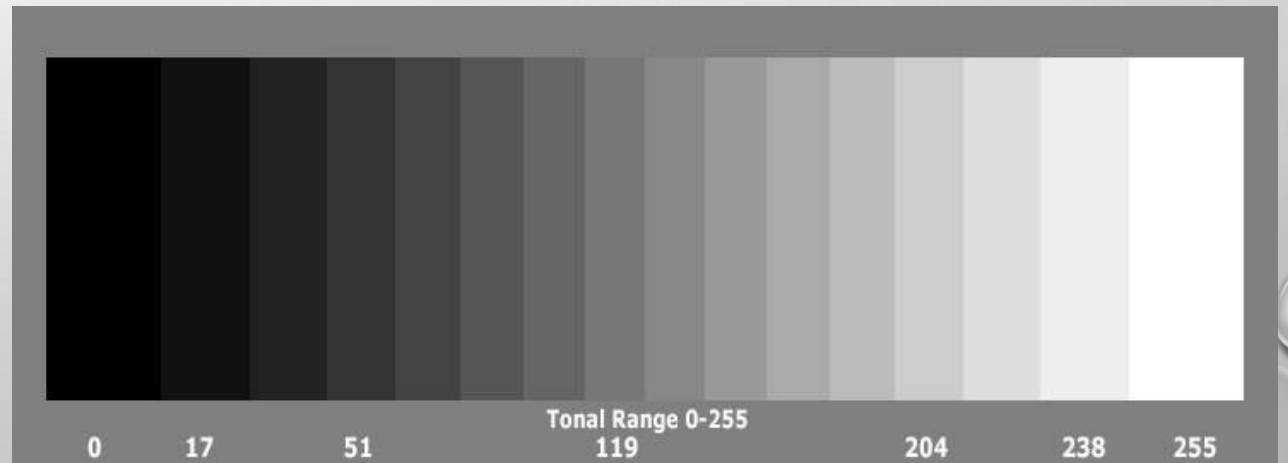
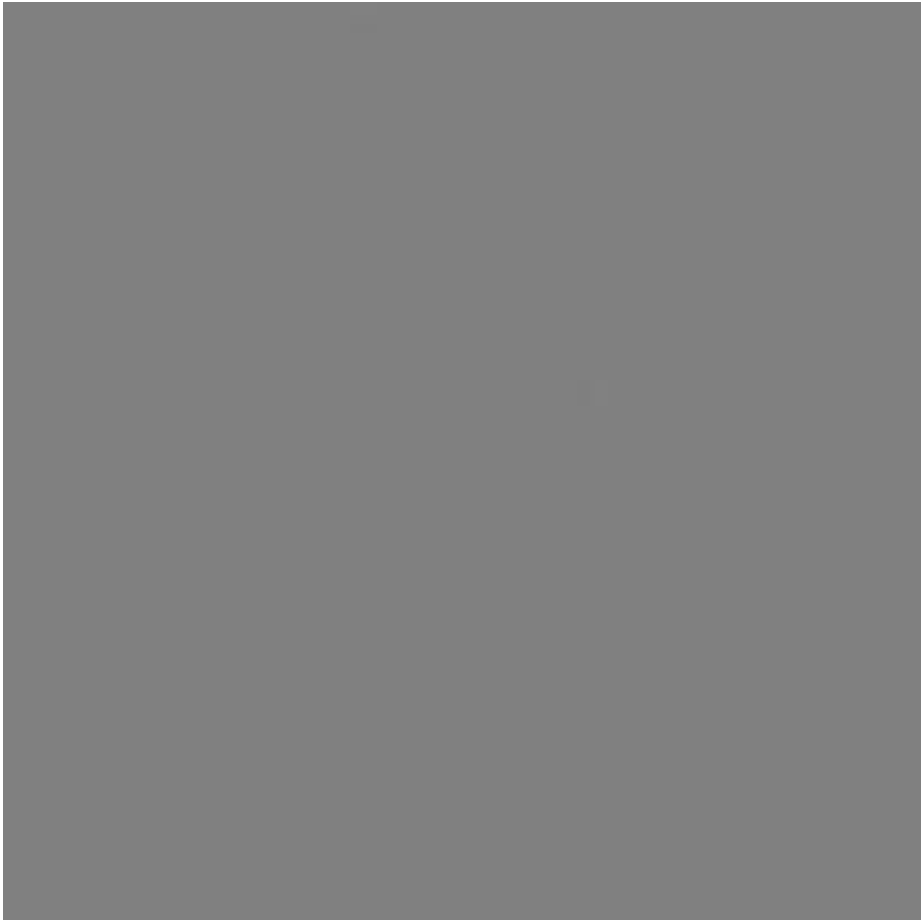
frame is moving w.r.t time (t)
refresh rate = 60 fps
At same coordinate (x, y) and diff times say $t_1, t_2, t_3, \dots$ that intensity remains same



- Interpixel Spatial redundancy is due to the correlation between the neighboring pixels in an image.
- That means neighboring pixels are not statistically independent. The gray levels are not equally probable.
- The value of any given pixel can be predicted from the value of its neighbors that is they are highly correlated.
- The information carried by individual pixel is relatively small. To reduce the interpixel Spatial redundancy the difference between adjacent pixels can be used to represent an image.

PSYCHOVISUAL REDUNDANCY (irrelevant information)

- Information that is not used by the human visual system are psychovisually redundant.
- The eye does not respond with equal sensitivity to all visual information.
- Certain information has less relative importance than other information in normal visual processing **psychovisually redundant** (which can be eliminated without significantly impairing the quality of image perception).
- The elimination of psychovisually redundant data results in a loss of quantitative information **lossy data compression method**.



CODING - DEFINITIONS

- ❑ **Code:** a list of symbols (letters, numbers, bits etc.)
- ❑ **Code word:** a sequence of symbols used to represent some information (e.g., Gray levels).
- ❑ **Code word length:** number of symbols in a code word.

Example: (binary code, symbols: 0,1, length: 3)

CODE CODE WORD

0: 000	4: 100
1: 001	5: 101
2: 010	6: 110
3: 011	7: 111

CODING REDUNDANCY

- ❑ $r_k \rightarrow$ Input Intensity Value e.g. 0 – 255 for grayscale image
- ❑ $l(r_k) \rightarrow$ No. of bits used to represent r_k
- ❑ Then average no of bits required to represent each pixel is

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$$

Say, uniform width str. (all pixels are using 8 bits to represent intensity)

$\rightarrow$ equal length code (fixed length code)

$\rightarrow$ m-bit fixed length code (8 bit fixed length code in our case)

CODING REDUNDANCY

❑ In this case, $l(r_k) = 8$ [each rq. 8 bits in fixed length]

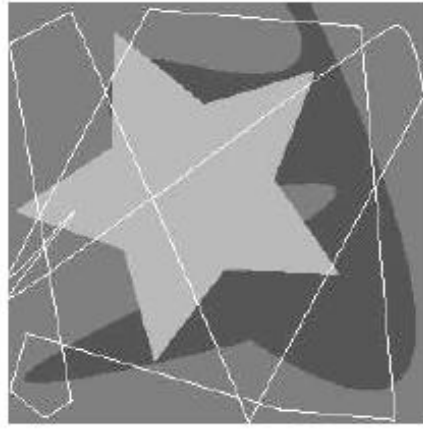
$$L_{avg} = 8 \sum_{k=0}^{L-1} p_r(r_k) = 8 * 1 = 8$$

❑ Coding redundancy tries to reduce L_{avg}

❑ Thus, total no. of bits required to represent an $M \times N$ image is MNL_{avg}

❑ The code can be an equal length code, or variable length code.

CODING REDUNDANCY



Code 1 ? equal length code
Code 2 ? variable length code
(obt by Huffman Coding)
Min bits ? higher probability

r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
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CODING REDUNDANCY

❑ For code 1, $L_{avg} = 8$

❑ On the other hand, using code 2, the average length of the encoded pixels is

$$L_{avg} = 0.25(2) + 0.47(1) + 0.25(3) + 0.03(3) = 1.81 \text{ bits}$$

❑ The resulting compression and corresponding relative redundancy are

Thus, 77.4% of the data in the original 8-bit 2-D intensity array is redundant.

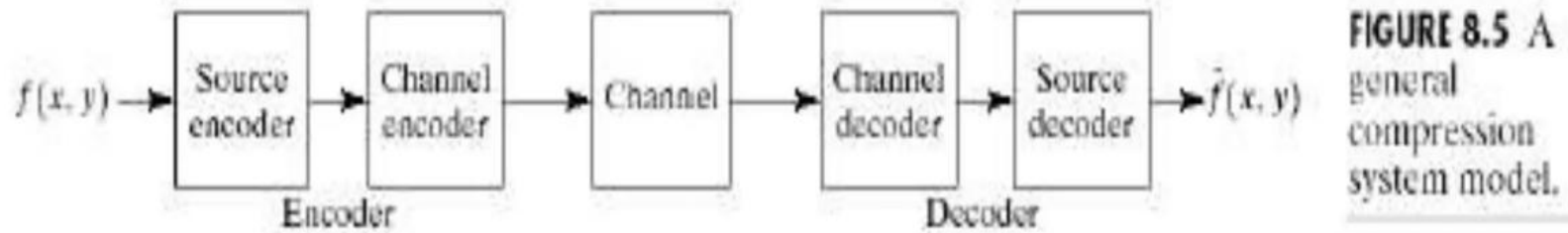
$$C = \frac{256 \times 256 \times 8}{256 \times 256 \times 1.81} \approx 4.42$$

$$R = 1 - \frac{1}{4.42} = 0.774$$

CODING REDUNDANCY

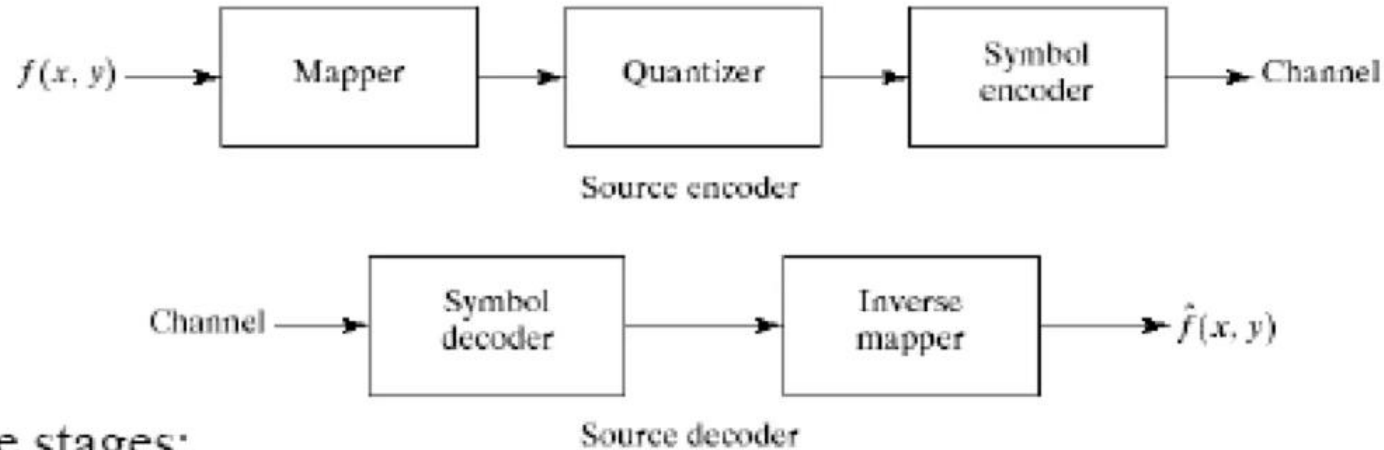
- ❑ Coding redundancy is present when the codes assigned to intensity values do not take full advantage of their probabilities of occurrence.
- ❑ The natural consequence is that, for most images, certain intensities are more probable than others.
- ❑ A natural binary encoding assigns the same number of bits to both the most and least probable values, failing to minimize and resulting in coding redundancy.

Image Compression Model



- The encoder creates a set of symbols (compressed) from the input data.
- The data is transmitted over the channel and is fed to decoder.
- The decoder reconstructs the output signal from the coded symbols.
- The source encoder removes the input redundancies, and the channel encoder increases the noise immunity.

Source Encoder and Decoder



Three stages:

Mapper: Transforms the input data into a (usually non-visual) format designed to reduce the interpixel redundancy in the image.

Quantizer: Reduces the accuracy of the mapper's output according to the predefined fidelity criterion. It is irreversible process and must be omitted in error-free compression scheme.

Symbol coder: Creates a fixed or variable length code to represent quantizer output, and maps the output in accordance with the code.

The background of the slide is a light gray gradient. In the top-left and bottom-right corners, there are several realistic-looking water droplets of various sizes, some overlapping. In the center of the slide, there is a faint, circular watermark logo. The logo features a globe with latitude and longitude lines, and the text "UNIVERSITY OF CALIFORNIA" is visible around the perimeter of the circle.

COMPRESSION METHODS

HUFFMAN CODING

- A measure to reduce coding redundancy
- Most popular coding redundancy technique
- Variable length code
- Min length code is assigned to one with highest probability

HUFFMAN CODING

Say, Image size: 10 x 10 (5 bit image)

Frequency:

a2 = 40 a6 = 30 a1 = 10 a4 = 10 a3 = 6 a5 = 4

$P(a2) = 40/100 = 0.4$

Symbols (like intensity levels)	Probabilities (sorted)	Source Reduction (do till two values are left) (Maintain in sorted order here as well)			
		1	2	3	4
a2	0.4	0.4	0.4	0.4	0.6
a6	0.3	0.3	0.3	0.3	0.4
a1	0.1	0.1	0.2	0.3	
a4	0.1	0.1	0.1		
a3	0.06	0.1			
a5	0.04				

Symbols (like intensity levels)	Probabilities (sorted)	Source Reduction (do till two values are left) (Maintain in sorted order here as well)			
		1	2	3	4
a2	0.4 1	0.4	0.4	0.4	→ 0.6 0
a6	0.3 00	0.3	0.3	0.3 00	→ 0.4 1
a1	0.1 011	0.1	→ 0.2 010	→ 0.3 01	
a4	0.1 0100	0.1 0100	0.1 011		
a3	0.06 01010	→ 0.1 0101			
a5	0.04 01011				

Encoded String: 010100111100

Decoding : a3 a1a2 a2 a6

Parameters:

- 1. Average length of code
 $L_{avg} = 0.4 * 1 + 0.3 * 2 + 0.1 * 3 + 0.1 * 4 + 0.06 * 5 + 0.04 * 5 = 2.2 \text{ bits/symbol}$
- 2. Total no. of bits to be transmitted
 $10 * 10 * 2.2 = 220 \text{ bits}$
- 3. How much you saved = $(10 * 10 * 5 - 10 * 10 * 2.2)/(10 * 10 * 5) = 0.56 = 56\%$

Resource:



resource